ICLAB 2025-Fall Lab09 Note

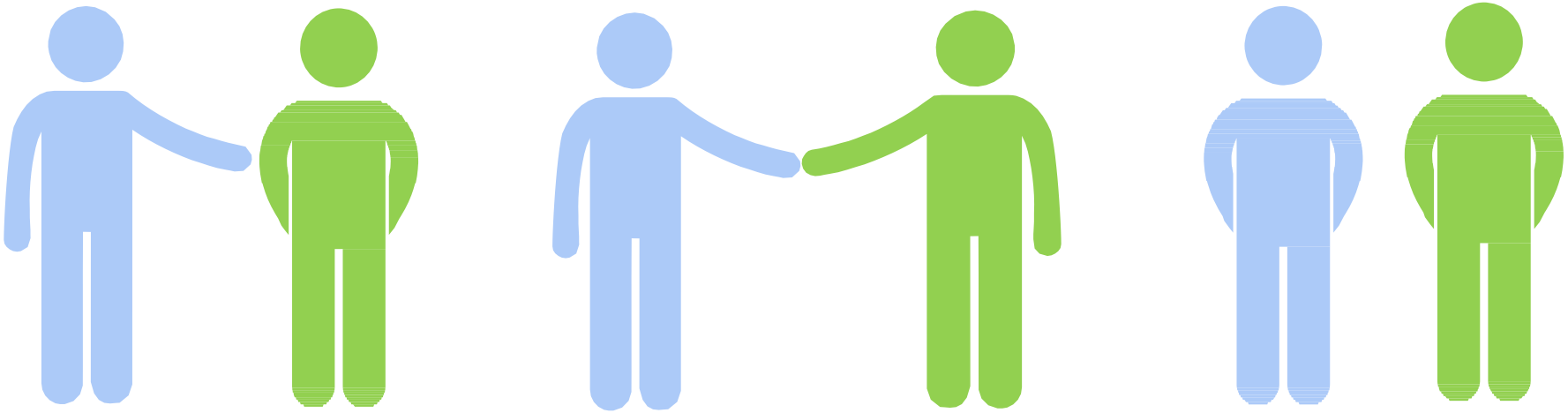
Outline

- **AXI4 Introduction**
  - Handshake Process
  - AXI Signal Description in Midterm Project
- **DRAM**
- **Reference Waveform**

AXI Handshake Process

Handshake Process Scenario 1

- **Slave: Valid** **Master: Ready**



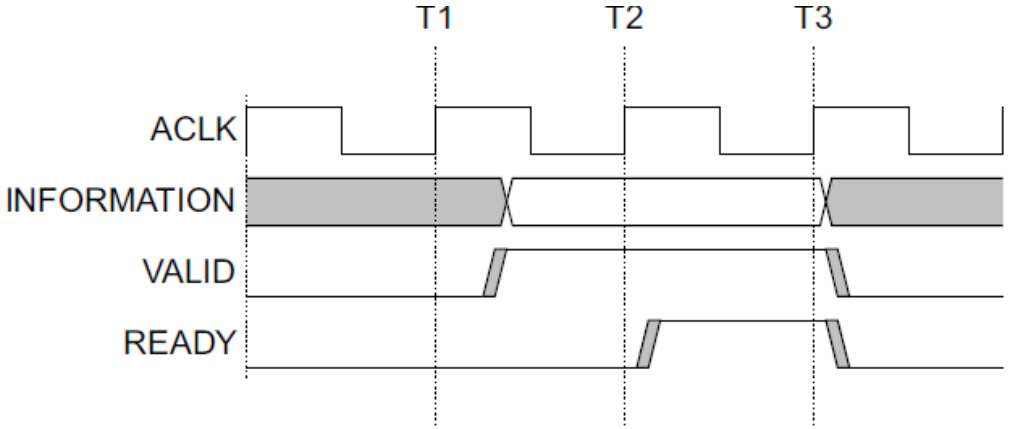
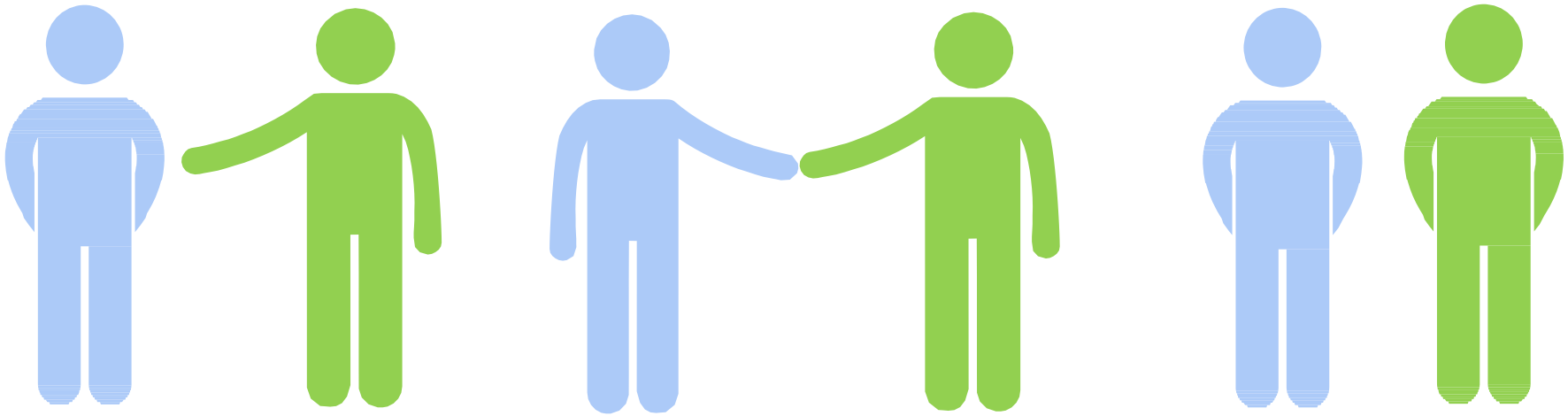


Figure A3-2 VALID before READY handshake

Handshake Process Scenario 2

- **Slave: Valid** **Master: Ready**



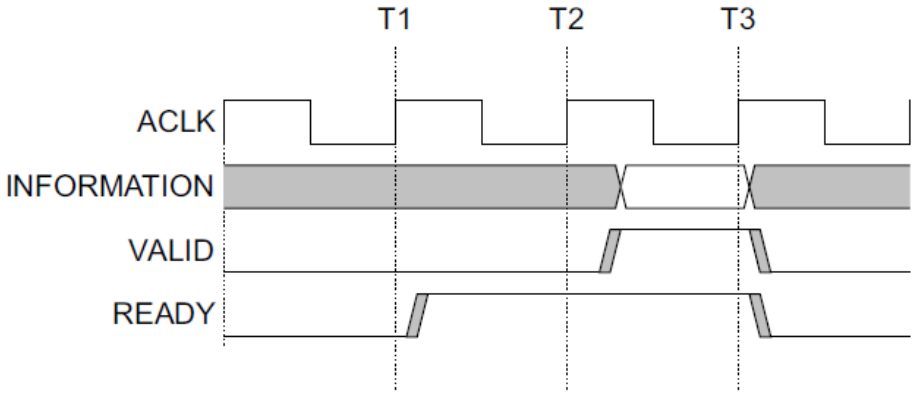
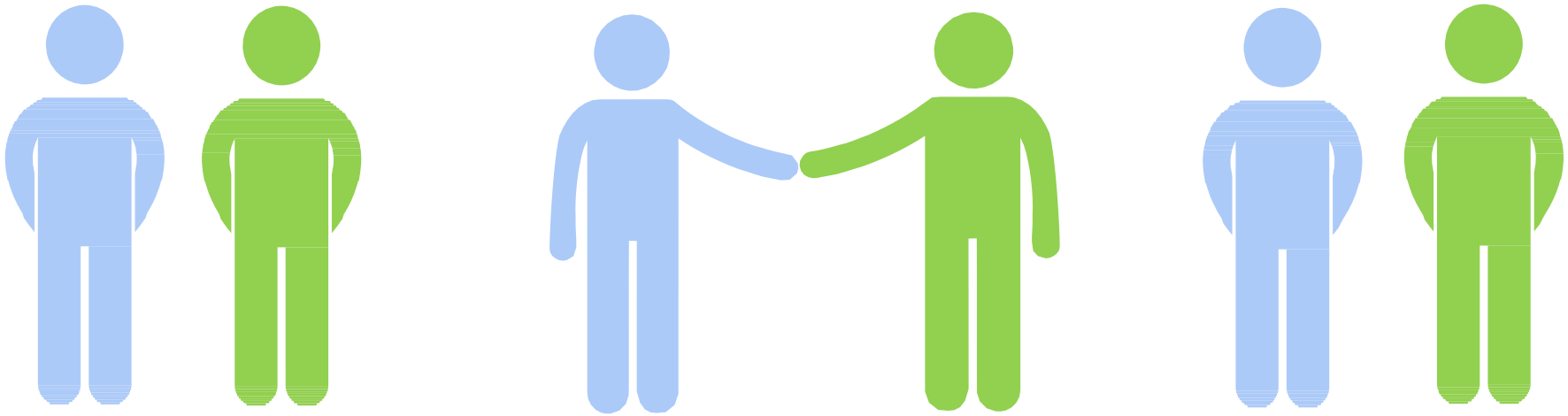


Figure A3-3 READY before VALID handshake

Handshake Process Scenario 3

- **Slave: Valid** **Master: Ready**



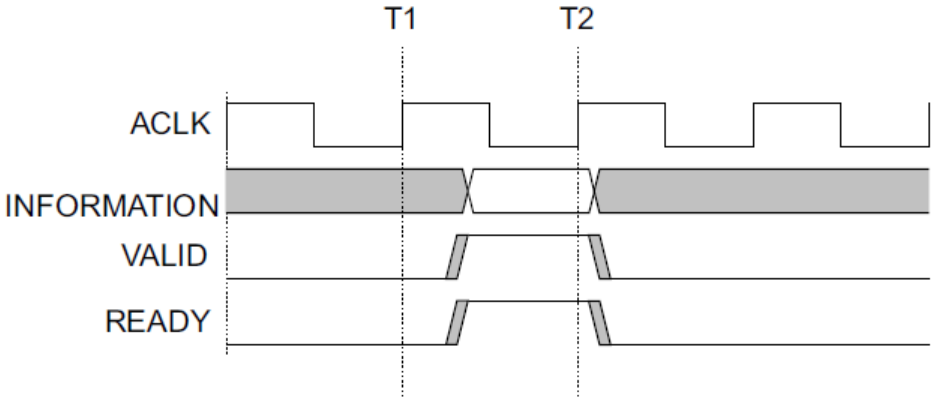


Figure A3-4 VALID with READY handshake

Handshake Process

- the VALID signal of one AXI component must not be dependent on the READY signal of the other component in the transaction.
- the READY signal can wait for assertion of the VALID signal.

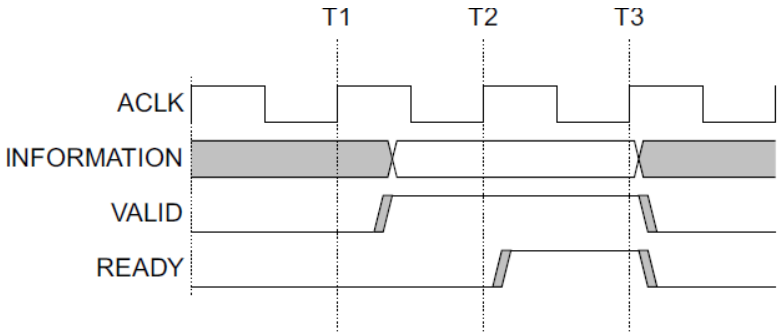


Figure A3-2 VALID before READY handshake

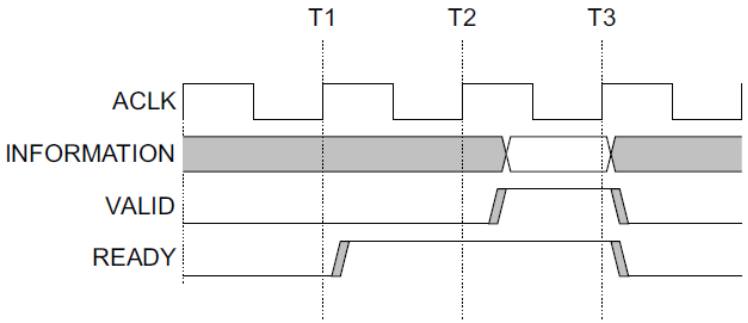


Figure A3-3 READY before VALID handshake

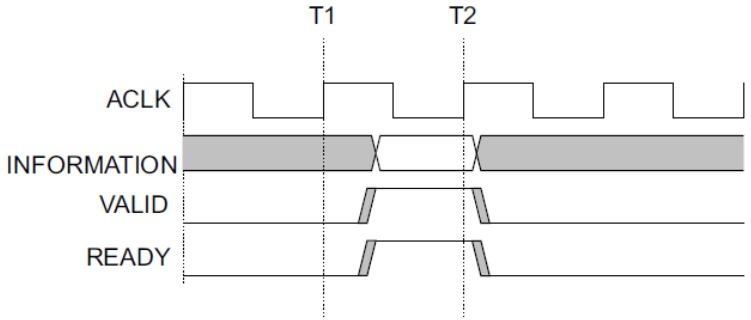


Figure A3-4 VALID with READY handshake

AXI4-Lite Introduction

AXI 4 Basic Transaction

Read Transaction

- Read Address Channel
- Read Data Channel

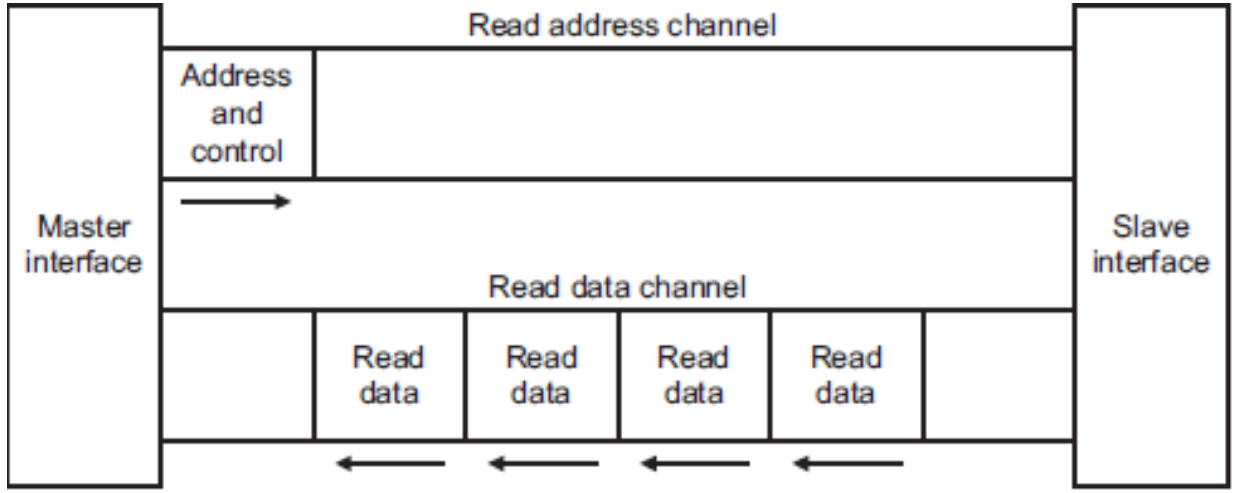


Figure 1-1 Channel architecture of reads

Each channel have valid-ready pair for handshaking process

Ref [2]

AXI 4 Basic Transaction

- **Write Transaction**
  - Write Address Channel
  - Write Data Channel
  - Write Response Channel

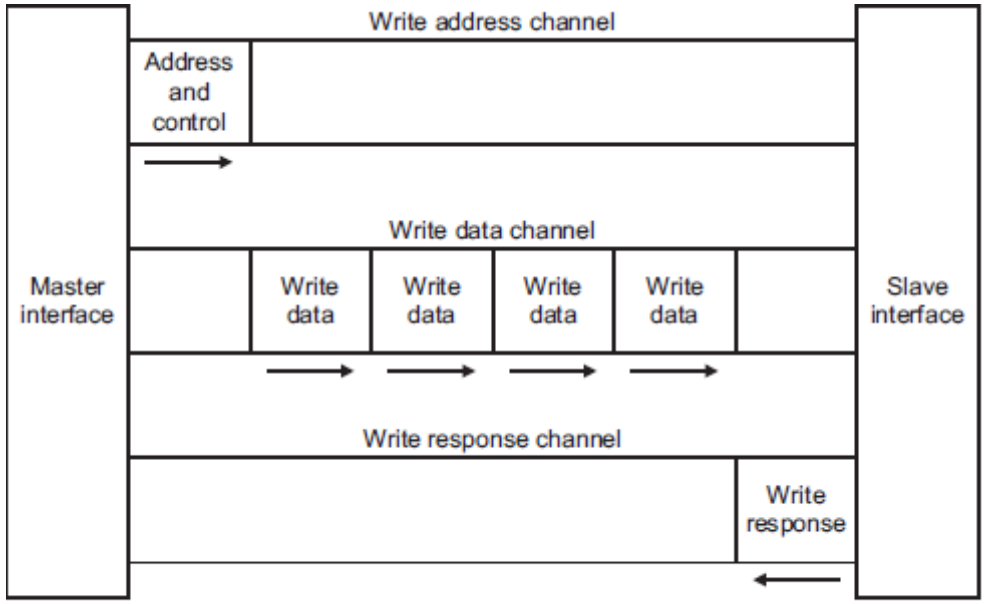


Figure 1-2 Channel architecture of writes

- **Each channel have valid-ready pair for handshaking process**

Ref [2]

Global Signals

Signal	Bit Width	Source	Description
clk	1	Clock source	Global clock signal. All signals are sampled on the **positive** edge of the global clock.
rst_n	1	Reset source	Global reset signal. This signal is active LOW.

Write Address Channel

Signal	Bit Width	Source	Description
AW_ADDR	32	Master	Write address.
AW_VALID	1	Master	Write address valid. This signal indicates that valid write address is available: 1 = address available 0 = address not available. The address remain stable until the address acknowledge signal, **AW_READY**, goes HIGH.
AW_READY	1	Slave	Write address ready. This signal indicates that the slave is ready to accept an address signals: 1 = slave ready 0 = slave not ready.

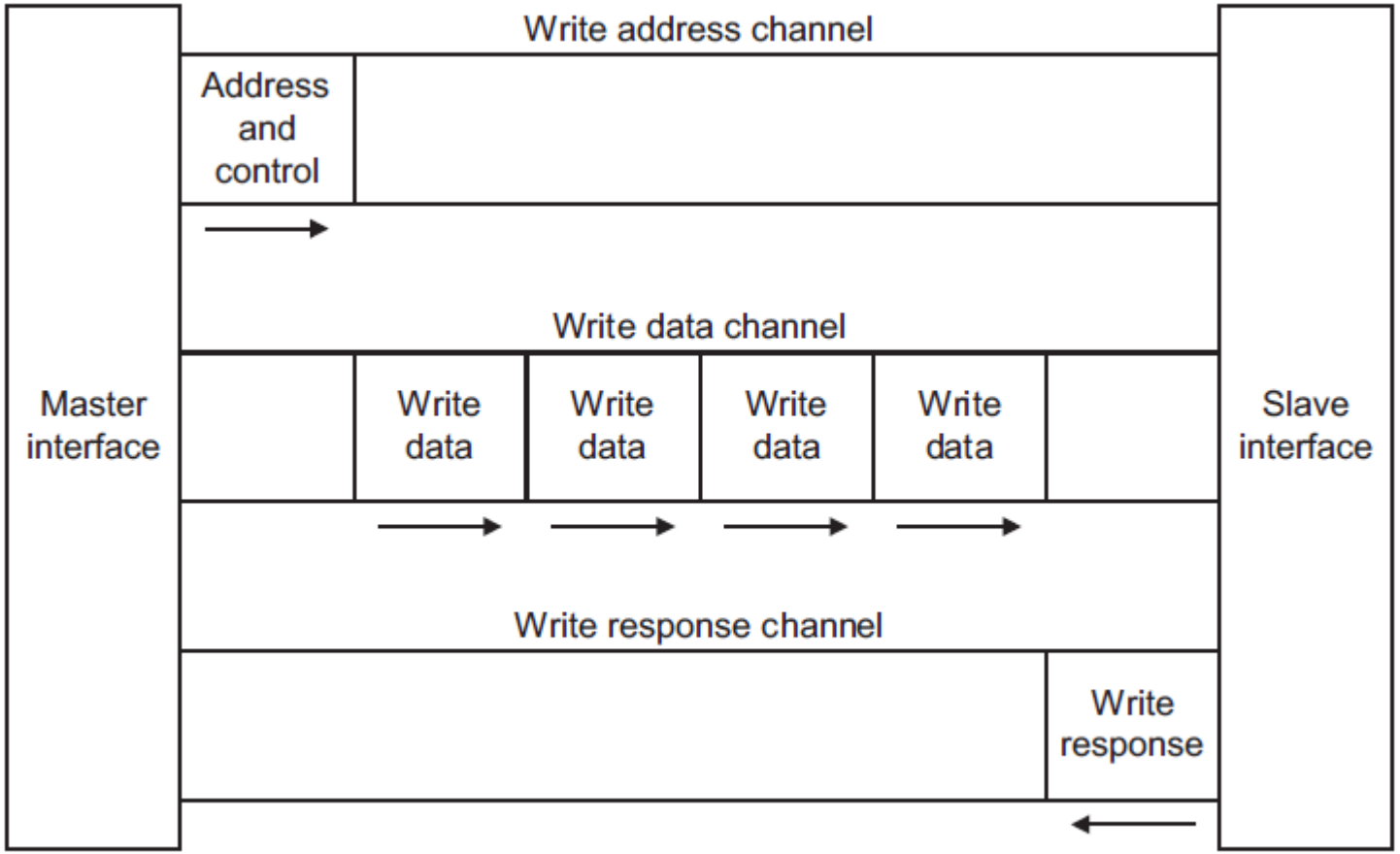
Write Data Channel

Signal	Bit Width	Source	Description
W_DATA	64	Master	Write data bus.
W_VALID	1	Master	Write valid. This signal indicates that valid write data is available: 1 = write data available 0 = write data not available.
W_READY	1	Slave	Write ready. This signal indicates that the slave can accept the write data: 1 = slave ready 0 = slave not ready.

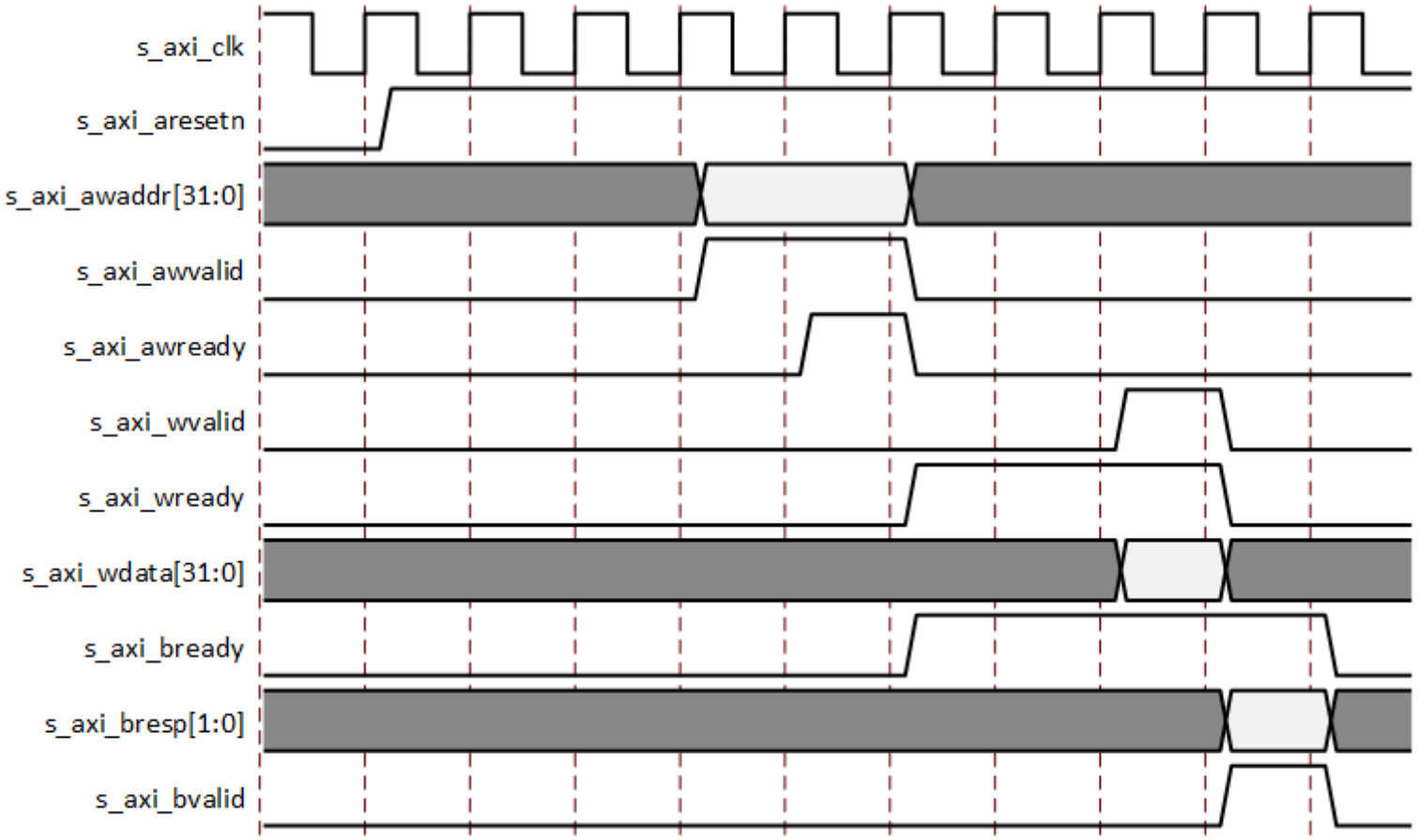
Write Response Channel

Signal	Bit Width	Source	Description
B_RESP	2	Slave	Write response. This signal indicates the status of the write transaction. The allowable responses are OKAY, EXOKAY, SLVERR, and DECERR. (In this lab we only issue OKAY(2'b00))
B_VALID	1	Slave	Write response valid. This signal indicates that a valid write response is available: 1 = write response available. 0 = write response not available.
B_READY	1	Master	Response ready. This signal indicates that the master can accept the response information. 1 = master ready. 0 = master not ready.

Write Transaction



Waveform of Write Transaction



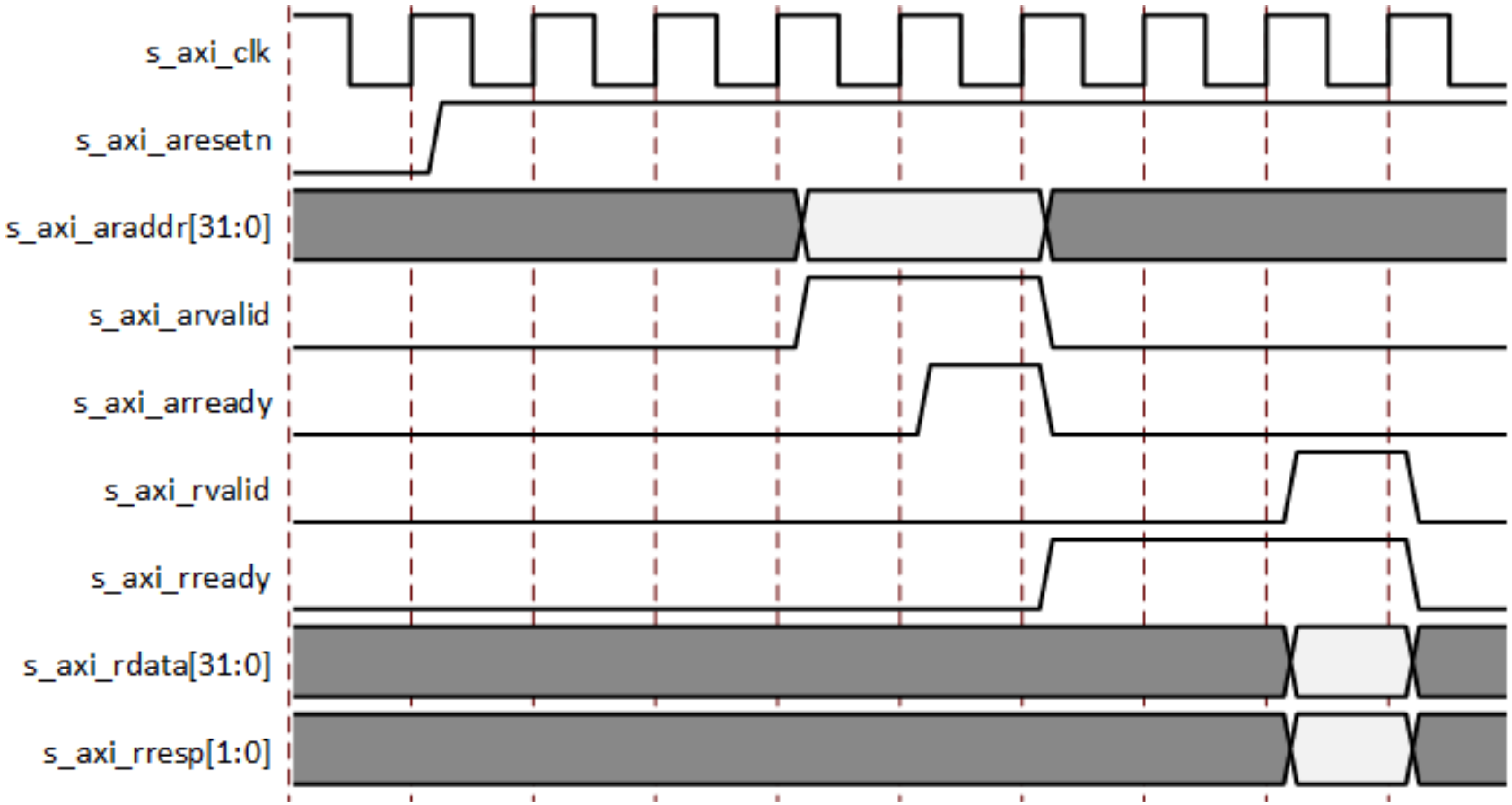
Read Address Channel

Signal	Bit Width	Source	Description
AR_ADDR	32	Master	Read address.
AR_VALID	1	Master	Read address valid. This signal indicates, when HIGH, that the read address is valid and will remain stable until the address acknowledge signal, **AR_READY**, is high. 1 = address and control information valid 0 = address and control information not valid.
AR_READY	1	Slave	Read address ready. This signal indicates that the slave is ready to accept an address signals: 1 = slave ready 0 = slave not ready.

Read Data Channel

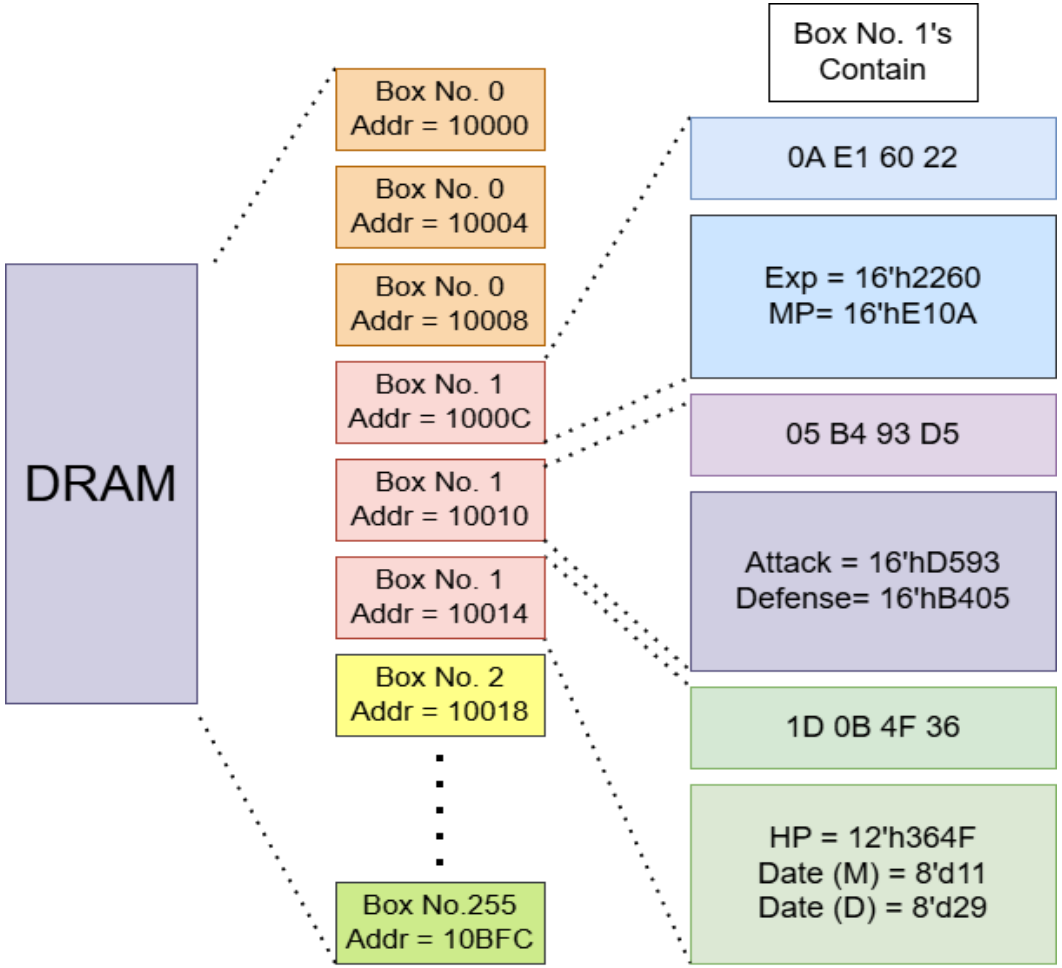
Signal	Bit Width	Source	Description
R_DATA	64	Slave	Read data.
R_RESP	2	Slave	Read response. This signal indicates the status of the read transfer. The allowable responses are OKAY, EXOKAY, SLVERR, and DECERR. (In this project we only issue OKAY)
R_VALID	1	Slave	Read valid. This signal indicates that the required read data is available, and the read transfer can complete: 1 = read data available 0 = read data not available.
R_READY	1	Master	Read ready. This signal indicates that the master can accept the read data and response information: 1= master ready 0 = master not ready.

Read Transaction



DRAM

DRAM structure



DRAM.dat file example

```
@10000
20 67 D7 E6
@10004
B4 11 A9 EC
@10008
17 06 75 4B
@1000C
A6 F2 33 40
@10010
FA 12 C9 77
@10014
09 08 4B FC
@10018
6B 53 2C 6A
@1001C
98 07 46 81
@10020
10 0A F1 1D
@10024
F8 58 05 5F
```

@1000C

A6 F2 33 40

@10010

FA 12 C9 77

@10014

09 08 4B FC

Category	Amount(Hex.)
Exp	4033
MP	F2A6
Attack	77C9
Defense	12FA
HP	FC4B
Date (Month)	08
Date (Day)	09

DRAM Note

- **You may modify the following part in ../00_TESTBED/pseudo_DRAM.sv**

DRAM latency →

```
parameter DRAM_R_latency = 1;
parameter DRAM_W_latency = 1;
parameter DRAM_B_latency = 1;
```

(Will be adjusted to 1~100 during DEMO)

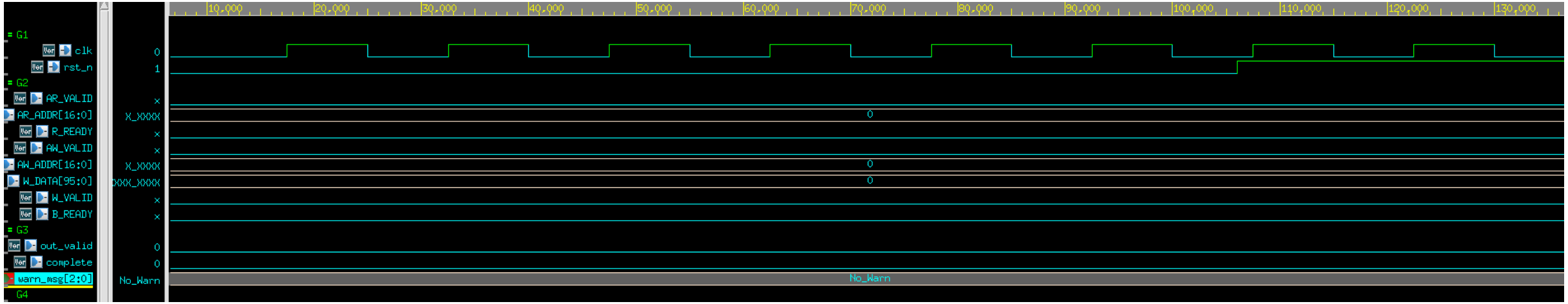
- **If you want to initialize dram in pattern, you may use the following code.**

Declaration of dram reg array→

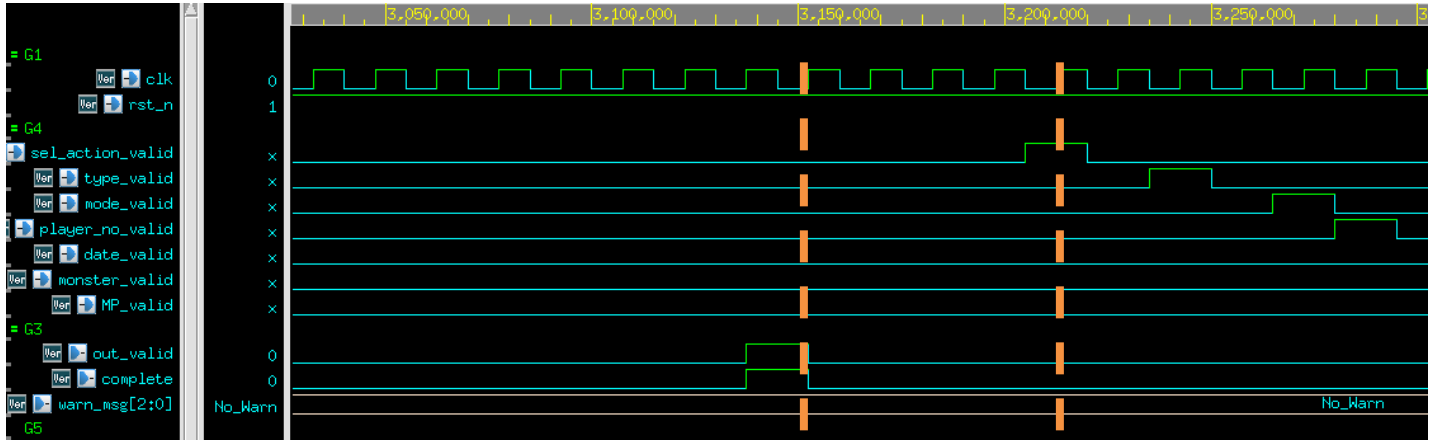
```
parameter DRAM_p_r = "../00_TESTBED/DRAM/dram.dat";

logic [7:0] golden_DRAM [((65536+12*256)-1):(65536+0)];

initial $readmemh(DRAM_p_r, golden_DRAM);
```

Waveform

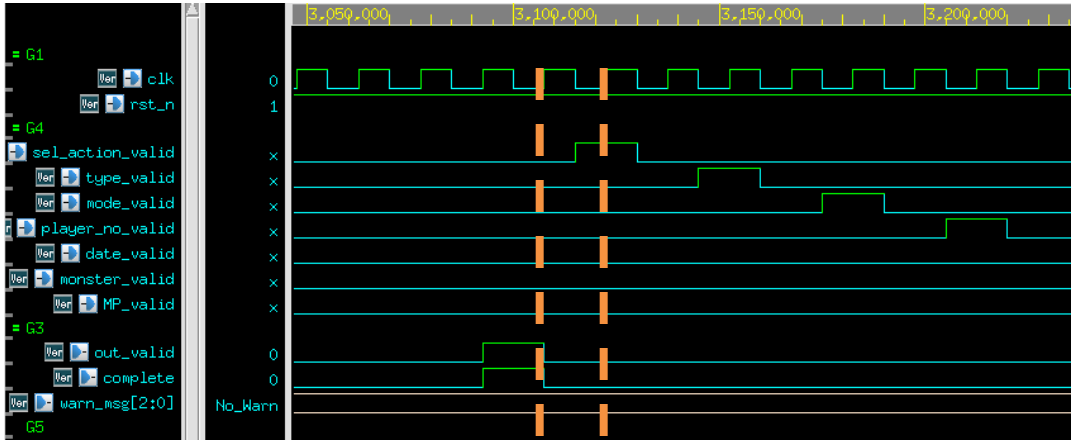
All output signals reset to 0 after rst_n



Start of the system



4 cycles



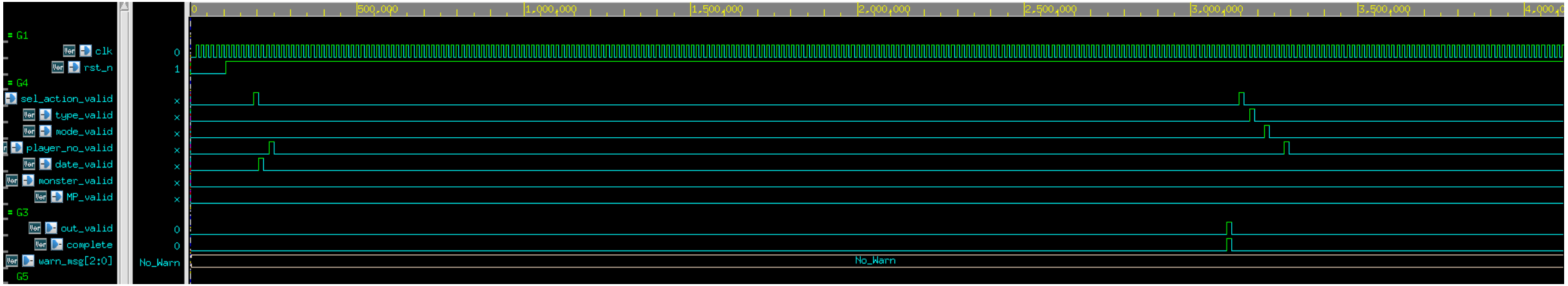
1 cycle

$1 \leq Gap \leq 4$

Order of input valid signal is fixed. Students can refer to the Lab9 Exercise operations at the page 1.

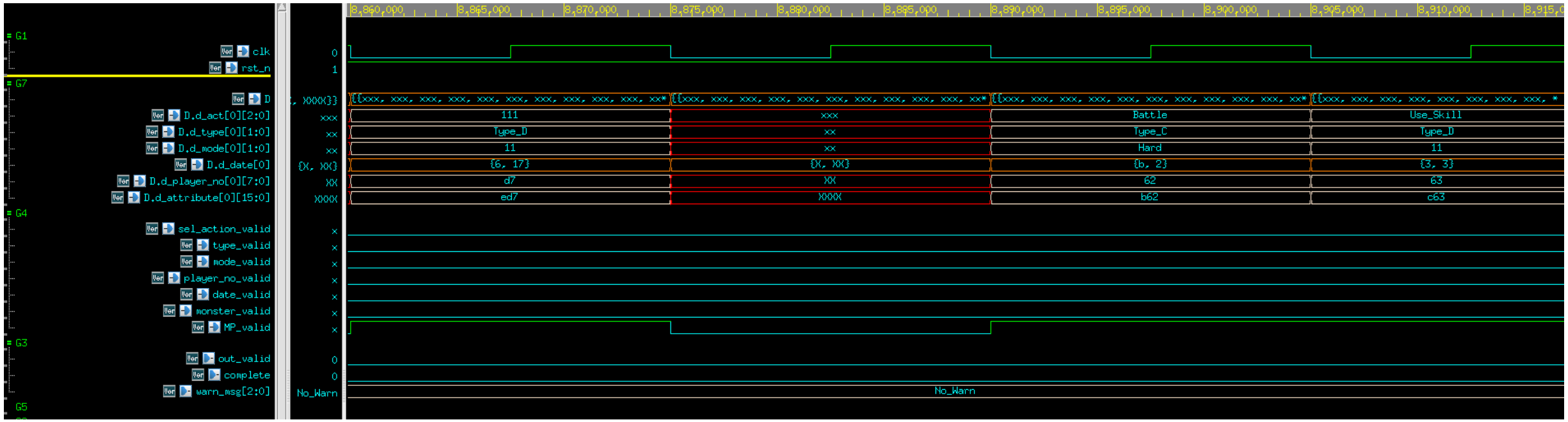
TA will not give the information of the delay between rst_n and the first input valid

Valid signal

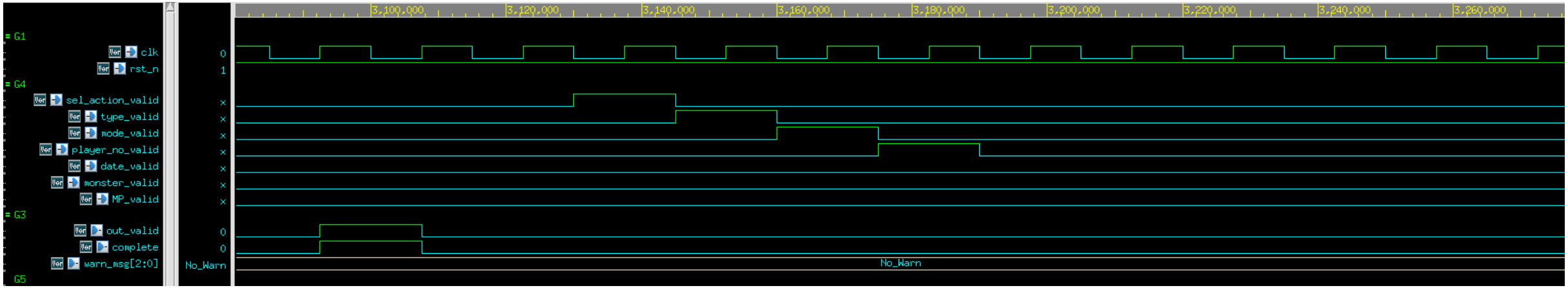
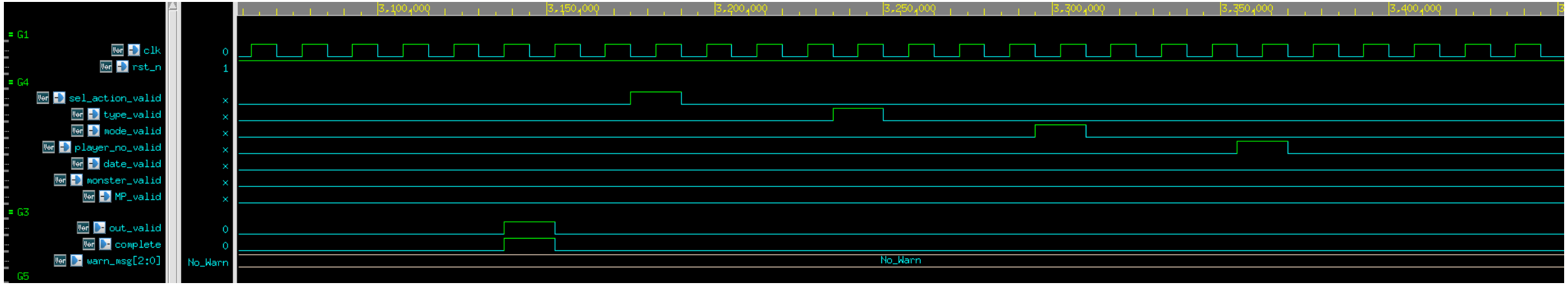


The next input valid signal among 7 input valid will become valid 1 to 4 cycles after a valid in 7 input valid or out_valid falls.

Monster_valid and MP_valid may or may not be continual



1 cycle between each valid signal



4 cycles between each valid signal